

1.2 & 1.3

Geographic Data

Objectives and Essential Learning

1.2 Identify different methods of geographic data collection.

- 1.2.1 Data may be gathered in the field by organizations or by individuals.
- 1.2.2 Geospatial technologies include geographic information systems (GIS), satellite navigation systems, remote sensing, and online mapping and visualization.
- 1.2.3 Spatial information can come from written accounts in the form of field observations, media reports, travel narratives, policy documents, personal interviews, landscape analysis, and photographic interpretation.



1.3 Explain the geographic effects of decisions made using geographical information.

- 1.3.1 Geospatial and geographical data, including census data and satellite imagery, are used at all scales for personal, business and organizational, and governmental decision-making purposes.

Identify different methods of geographic data collection.

1. How is data gathered?

- a. Fieldwork/Field Observations: The act of an individual physically visiting a location or place and recording, firsthand, information there.
 - i. Examples: Written accounts, travel narratives, media reports, policy documents, interviews, photographs, landscape analysis



FIELDWORK COLLECTION & TECHNIQUES

Geographical enquiry encourages questioning, investigation and critical thinking about issues affecting the world and people's lives, now and in the future. Fieldwork is an essential element of good geographical work and helps to bring the information to life in a way that other activities find impossible to do.

FIELDWORK INVOLVES...

Asking geographical questions, thinking critically, constructively and creatively

Planning geographical enquiries, suggesting appropriate sequences of investigation

Collecting and recording data

Identifying bias, opinion and abuse of evidence in sources when investigating issues

Presenting data in an accurate and meaningful way

Analysing and evaluating evidence collected from all sources

Coming to conclusions based on the data

Evaluating your methods and techniques

Identify different methods of geographic data collection.

Another example of fieldwork - The US Census

- Mandated by the Constitution
- Ex. of an organization (US government) doing fieldwork.

What is the point?

- a. Representative Districts - voting & democracy
- b. Distribution of public funding for:
 - i. Schools
 - ii. Law Enforcement & Fire Department
 - iii. Highways & Infrastructure
- c. Personal, government & business decision making.



Identify different methods of geographic data collection.

b. Geospatial Technologies

- Geographic information systems (GIS)
- Satellite navigation systems (GPS)
- Remote sensing
- Online mapping + visualizations

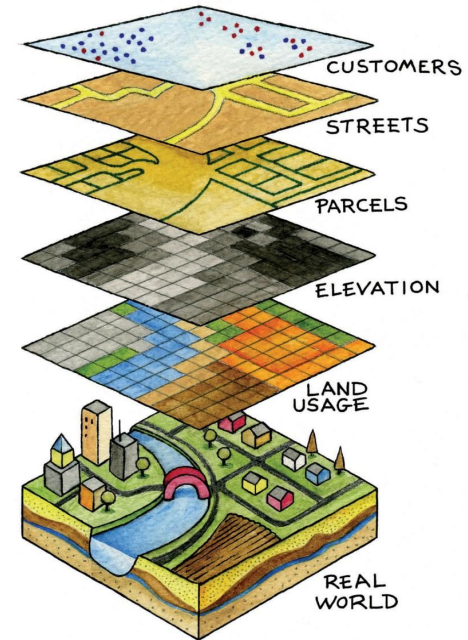
Geospatial technologies are utilized by businesses, organizations, individuals and government agencies to make decisions.



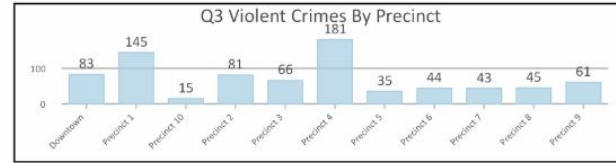
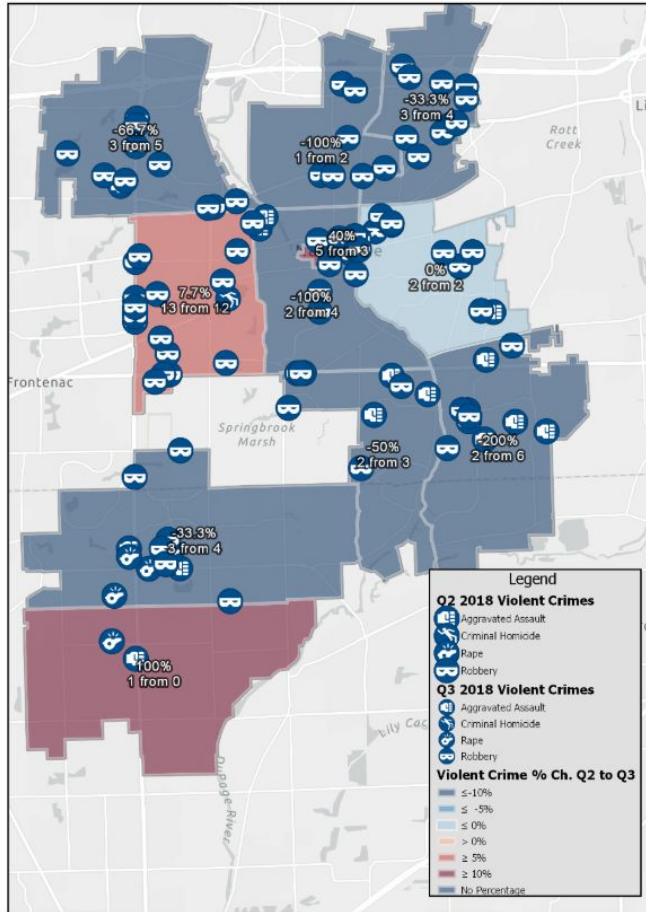
Identify different methods of geographic data collection.

Geographic Information Systems (GIS)

- Computer system/software that stores, analyzes, and displays information from multiple digital maps or data sets.
- Thematic and physical layers
- Could be used for...
 - Analysis of crime data
 - Effects of pollution
 - Urban planning



Violent Crime By Precinct (Percent Change Q2-Q3)



Offense ID	Reported Date	Full Address	UCR Type	Weapon	Case Status	Precinct
2010015553	8/23/2018 1:37:01 AM	1305 Mermouth Av	Aggravated Assault	Firearm	Warrant Advised	Precinct 8
2010009956	9/21/2018 10:42:00 AM	2511 W 75th St	Clear		Clear	Precinct 4
2010007577	9/20/2018 7:34:01 AM	2035 Temple Dr	Robbery		Clear	Precinct 7
2010012346	9/20/2018 6:33:01 AM	2220 83rd St	Robbery		Clear	Precinct 5
2010002213	9/18/2018 7:35:00 AM	892 W 75th St	Robbery		Clear	Precinct 7
2010000036	9/17/2018 5:12:00 AM	3032 English Rows Av	Aggravated Assault		Clear	Precinct 10
2010003533	9/16/2018 4:51:00 AM	1295 E Ogden Av	Aggravated Assault	Firearm	Active	Precinct 3
2010007738	9/16/2018 12:35:00 AM	125 S Washington St	Robbery	Firearm	Clear	Downtown
2010002041	9/14/2018 5:17:00 PM	400 S Main St	Robbery	Firearm	Clear	Downtown
2010010177	9/14/2018 1:12:00 PM	2775 Showcase Dr	Aggravated Assault		Warrant Advised	Precinct 9
2010011204	9/13/2018 3:00:00 PM	2780 Fitness Dr	Robbery	Knife	Unfounded	Precinct 4
2010007168	9/13/2018 4:36:00 AM	2808 Kilbourne Ln	Robbery	Knife	Clear	Precinct 9
2010011116	9/13/2018 2:05:00 AM	22 E Chicago Av	Robbery	Knife	Clear	Downtown
2010011562	9/13/2018 1:58:00 AM	1003 N Harper St	Robbery		Clear	Precinct 3
2010007184	9/12/2018 6:43:00 AM	1403 Lawrence Dr	Robbery	Knife	Active	Precinct 9
2010001870	9/11/2018 3:00:00 PM	610 W Jefferson Av	Robbery		Clear	Downtown
2010011401	9/11/2018 12:42:00 PM	2780 Fitness Dr	Robbery	Firearm	Clear	Precinct 4
2010008784	9/11/2018 5:08:00 AM	3159 Kingsford Ln	Rape		Clear	Precinct 9
2010011514	9/10/2018 8:32:01 AM	680 S Route 55	Robbery	Knife	Active	Precinct 4
2010000014	9/9/2018 1:40:00 AM	117 S Eagle St	Robbery		Clear	Downtown
2010006333	9/6/2018 1:03:00 AM	440 Aurora St	Robbery		Clear	Downtown
2010011333	9/6/2018 4:39:00 AM	140 N Wright St	Robbery		Clear	Precinct 6
2010006873	9/5/2018 3:05:00 AM	552 N Loomis St	Robbery	Knife	Clear	Precinct 2
2010011961	9/1/2018 7:19:00 AM	2855 95th St	Rape		Clear	Precinct 9
2010016147	8/27/2018 7:27:00 AM	552 S Route 55	Robbery	Firearm	Active	Precinct 4
2010000774	8/24/2018 3:00:00 PM	405 W Jefferson Av	Robbery		Clear	Precinct 5
2010044170	8/23/2018 5:03:01 AM	2815 Hawthorne Ln	Rape		Warrant Advised	Precinct 10
2010001266	8/23/2018 5:55:00 AM	1532 North Aurora Rd	Clear		Clear	Precinct 1
2010006645	8/24/2018 3:07:01 PM	720 E Bowser Rd	Robbery		Clear	Downtown
2010006936	8/24/2018 7:00:00 AM	14 Jackson Av	Robbery	Firearm	Clear	Precinct 3
2010005777	8/19/2018 10:06:01 AM	2780 Fitness Dr	Robbery		Clear	Precinct 4
2010007386	8/19/2018 8:42:00 AM	30 N Brainerd St	Robbery		Clear	Precinct 6
2010011585	8/19/2018 2:23:00 AM	22 E Chicago Av	Robbery	Firearm	Clear	Downtown
2010007703	8/18/2018 7:24:00 AM	1326 N Route 55	Robbery	Knife	Clear	Precinct 1
2010011820	8/17/2018 10:36:01 AM	1320 S Route 55	Robbery		Active	Precinct 9
2010007073	8/17/2018 12:23:01 AM	126 N Wright St	Robbery		Clear	Precinct 6
2010016344	8/12/2018 2:59:00 AM	22 E Chicago Av	Robbery		Clear	Downtown
2010007950	8/9/2018 8:56:00 AM	794 Island Ct	Aggravated Assault	Firearm	Warrant Advised	Precinct 1
2010013126	8/4/2018 11:44:00 PM	2780 Fitness Dr	Aggravated Assault		Clear	Precinct 4
2010016353	7/30/2018 5:45:00 AM	4502 Ogden Av	Robbery		Clear	Precinct 3
2010001787	7/25/2018 3:00:00 PM	1437 N Eagle St	Robbery		Active	Precinct 2
2010007462	7/25/2018 4:44:00 AM	1408 W Jefferson Av	Robbery		Clear	Precinct 4
2010011879	7/23/2018 8:17:00 AM	1163 E Ogden Av	Robbery	Knife	Clear	Precinct 3
2010009852	7/16/2018 2:03:01 AM	1119 Essex Ct	Robbery	Knife	Clear	Precinct 4
2010001565	7/15/2018 3:00:00 PM	735 S West St	Robbery		Clear	Precinct 5
2010019558	7/15/2018 10:34:01 AM	424 E Bailey Rd	Robbery		Clear	Precinct 8
2010007786	7/13/2018 8:21:00 AM	838 Broadway Dr	Robbery		Clear	Precinct 4
2010010794	7/11/2018 8:47:00 AM	2707 Aurora Av	Robbery		Clear	Precinct 4
2010005600	7/6/2018 11:48:01 AM	860 W 75th St	Aggravated Assault		Warrant Advised	Precinct 7
2010002685	7/6/2018 8:58:00 AM	121 S Washington St	Aggravated Assault	Firearm	Clear	Downtown
2010000087	7/5/2018 7:52:00 AM	1520 Riparian Dr	Aggravated Assault	Blunt Object	Warrant Advised	Precinct 7
2010000038	7/2/2018 2:44:00 PM	1592 W Jefferson Av	Robbery		Clear	Precinct 4
2010000787	7/2/2018 12:55:00 AM	454 S Route 55	Robbery	Firearm	Clear	Precinct 4
2010014574	7/1/2018 6:14:00 AM	2300 95th St	Aggravated Assault		Clear	Precinct 9

Prepared by Chris Delaney on 10/2/18 for CompStat period 24.
Geocoding Rate 99% (Q2 n = 135, Q3 n = 121).



Identify different methods of geographic data collection.

Satellite Navigation Systems/Global Positioning System (GPS)

- Satellites orbit the earth and communicate locational information to GPS receivers.
- Absolute location
 - Navigation -> ships, cars, aircraft

Aerial Photography

- Professional images captured from planes or drones.



- ■ In response to the deadly and devastating tornadoes in Tennessee, our fixed-wing aircraft was deployed to
- ■ capture high-resolution imagery of the tornado's devastating path. Overnight, images were sent to be
- ■ processed, and the first look at the damage provided crucial information to aid recovery efforts within 27 hours of responding to the event.



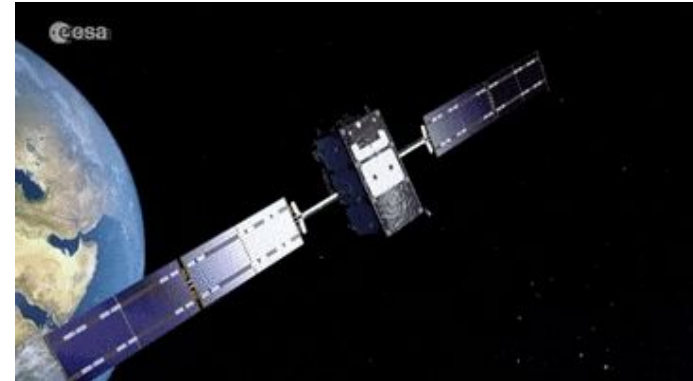


Pre-event and post-event imagery help indicate the severity of the damage

Identify different methods of geographic data collection.

Remote Sensing

- The use of cameras or other sensors mounted on aircraft or satellites which orbit the earth above the atmosphere to collect digital images of the earth's surface.
- Examples:
 - Monitoring environmental changes
 - Weather



Example of Remote Sensing: North Korean Prison Camps

"We worked in the farms (at kwanliso 15) from 7am to 8pm. We cultivated corn. We were divided to work in units comprising 10-15 people each. We were given a daily production target that we had to meet. If the unit did not meet the daily target, the unit-members were punished collectively. During the course of our three-year detention, often we did not meet our targets because we were always hungry and weak. We were punished with beatings and also reductions in our food quota. In addition to that, in the Ideology Struggle Sessions that were held after work, those who did not meet the target were severely criticized and beaten by other inmates."



DigitalGlobe Natural Color Imagery, [June 11, 2010](#), 41.261302°, 129.375145°

In order to circumvent the unwillingness of the North Korean authorities to allow human rights investigators access, Amnesty International and others have used remote sensing tools, such as satellite imagery, to provide more information about the human rights situation in North Korea. Previous satellite image analysis released by Amnesty International suggests the use and expansion of North Korea's notorious political prison camps, and the blurring of the boundary between kwanliso 14 and surrounding villages.

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